Manual calculation

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1. Program A: 8 BIT Addition and Subtraction

A = 5AH, B = 27H

Addition

5AH + 27H = 81H

Sum = 81H, CY = 0

Ac = 1, P = 1, Z = 0, S = 1

Flag byte = 94H

Signed overflow = Yes

Subtraction:

5AH − 27H = 33H

Difference = 33H, Borrow = 0

Ac = 0, P = 1, Z = 0, S = 0

Flag byte = 06H

Signed overflow = No.

2. Program B: 16 - Bit Addition and Subtraction

Given X = 1234H Y = 4321H

Addition:

1234H + 4321H = 5555H

Carry = 0

Subtraction:

1234H − 4321H = CF13H

Borrow = 1

3. Program C: 8 bit Multiplication and Division.

Given Multiplication 14H X 06H.

Division : F2H ÷ 0AH

Multiplication :

14H X 06H = 0078 H

Product = 0078H

Division :

F2H ÷ 0AH

Quotient = 18 H

Remainder = 02 H

F2H = (18H X 0AH) + 02H

4. Program D: 16 bit Multiplication and Division

Multiplication:

1357 H X 0025H = 0002CA13H

Upper Word = 0002 H

Lower Word = CA13 H

Division

9A32H : 0011 H

Quotient = 0911 H

Remainder = 0002H

9A23H = (0911H X 0011 H) + 0002H.